Range Resolution Enhanced Method With Spectral Properties for Hyperspectral LiDAR

Yuhao Xia, Shilong Xu, Hui Shao, Ahui Hou, Jiajie Fang, Fei Han, Youlong Chen, Jiaqi Wen, Fashuai Li, Yuwei Chen, and Yihua Hu

***Abstract*— Waveform decomposition is needed as a first step in the extraction of various types of geometric and spectral information from hyperspectral full-waveform light detection and ranging (LiDAR) echoes. We present a new approach to deal with the "pseudo-monopulse" waveform formed by the overlapped waveforms from multitargets when they are very close. We use one single skew-normal distribution (SND) model to fit waveforms of all spectral channels first and count the geometric center position distribution of the echoes to decide whether it contains multitargets. The geometric center position distribution of the "pseudo-monopulse" presents aggregation and asymmetry with the change of wavelength, while such an asymmetric phenomenon cannot be found from the echoes of the single target. Both theoretical and experimental data verify the point. Based on such observation, we further propose a hyperspectral waveform decomposition method utilizing the SND mixture model with: 1) initializing new waveform component parameters and their ranges based on the distinction of the three characteristics (geometric center position, pulsewidth, and skew coefficient) between the echo and fit SND waveform; 2) conducting single-channel waveform decomposition (SCWD) for all channels; 3) setting thresholds to find outlier channels based on statistical parameters of all single-channel decomposition results (the standard deviation and the means of geometric center position); and 4) reconducting SCWD for these outlier channels. The proposed method significantly improves the range resolution from 60 to 5 cm at most for a 4-ns width laser pulse and represents the state-of-the-art in "pseudo-monopulse" waveform decomposition.**

***Index Terms*— Full-waveform light detection and ranging (LiDAR), hyperspectral LiDAR (HSL), range resolution, waveform decomposition.**

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I. Introduction

NOWADAYS, full-waveform light detection and ranging (FWL) is a widely used, mature, and popular active detection technology for remote sensing [1], [2], [3]. By processing full-waveform light detection and ranging (LiDAR) data, various information can be retrieved, such as target count, range, and inclination. These features can be used for 3-D point cloud expansion and identification, biomass estimation, vegetation detection, urban terrain modeling, and other remote sensing transactions [4], [5], [6], [7], [8]. With the emergence of multispectral LiDAR (MSL) and hyperspectral LiDAR (HSL) [9], [10], [11], the generated spectral 3-D cloud has a significant impact on remote sensing and other fields where target 3-D detection and identification is crucial [10], [12], [13], such as civil engineering, cultural heritage preservation, material processing, or geomorphological studies [14], [15], [16].

There is plenty of research on single-wavelength waveform decomposition methods, which can be divided into direct decomposition and deconvolution. For direct decomposition, researchers introduced various mathematical models, such as Gaussian, Burr, skew-normal distribution (SND), and B-spline, to fit the echo and extract waveform component parameters [12], [17], [18], [19], [20], [21], [22]. However, the deconvolution performs better than the direct decomposition in revealing overlapping, hidden, and weak response waveform components. Other researchers proposed the Wiener filtering, regularization, Richardson–Lucy (RL), Gold, and other methods in detecting overlapped subwaveforms [23], [24], [25], [26], [27].

When introducing spectral factor in full-waveform light detection and ranging (FWL), Wang et al. [13] proposed a multichannel interconnection method for HSL by stacking waveforms with a high signal-to-noise ratio (SNR) to reveal hidden components. They improved the range resolution from 60 to 43 cm corresponding to the laser pulsewidth of 4 ns and at a distance of 500 m. Xia et al. [28] proposed a 101-channel HSL waveform decomposition method by looking for outliers in different channels and helping them redecompose. Also, they improved the range resolution of the laser signal with 4-ns pulsewidth to 45 cm.

Single-target echo will be broadened due to the change of projection of laser footprint. When two targets get very close, their overlapped echoes form a "pseudo-monopulse"

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waveform, which is very similar to the single target's broadened echo. The pictures in Fig. 1 describe the two situations. However, the current waveform decomposition method will treat these overlapped waveforms as the case of the single target's broadened echo.

Currently, for the single-wavelength waveform direct decomposition method, when the overlap rate of two echoes is greater than 60%, the number of subwaveforms cannot be effectively estimated by inflection points or other methods [24]. redFor deconvolution, the prevailing RL [24] cannot deal with the situation when the overlap rate reaches 90% [24]. The robust deconvolution method proposed by Liu et al. [24] can overcome the disadvantage of RL, enabling that the laser pulse whose width is 8.7 ns distinguishes two targets with a neighbor distance (ND) of 120 cm [29]. However, the results may not be reliable because the "pseudo-monopulse" waveforms and the single target's broadened waveforms are not distinguished in advance, and the target discrimination/differentiation capability is limited by the width of the transmitted pulse. The current multispectral [30] and hyperspectral [13], [28] waveform decomposition method cannot deal with the "pseudo-monopulse" either if no significant echo deformation features appear.

To process the "pseudo-monopulse" and further improve the FWL's range resolution, in this article, we find one critical and traceable statistical phenomenon for hyperspectral measurements of HSL. More specifically, the finding is that the distribution of geometric center position of all spectral echoes of the "pseudo-monopulse" shows different statistical distribution compared with that of single target. The geometric center position distribution of the "pseudo-monopulse" presents aggregation and asymmetry, while that of single targets shows a more random distribution. Based on such observation, we propose a method to efficiently judge the waveform types and a decomposition method to reveal the overlapped subwaveforms, even when they are extremely close (5 cm). The main contributions of this study can be itemed as follows.

1) We find that the distribution of geometric center position of the "pseudo-monopulse" waveform shows aggregation and asymmetry in the spectral domain, which is significantly different from that of a single-target echo waveform.
2) We propose an effective waveform-type decision method and further design a method to decompose the "pseudo-monopulse" waveform, including a special initial parameter estimation method and a hyperspectral waveform decomposition method.
3) A range resolution experiment is conducted to prove and evaluate the effectiveness, robustness, and feasibility of the waveform-type decision method and examine the accuracy and performance of our waveform decomposition method.

The remainder of this article is organized as follows. Section II describes the derivation for the physical phenomenon by theoretical modeling and explains the proposed method in detail. Section III describes the full-waveform HSL (FWHSL) system and experimental design prepared for the verification of the theory and proposed method. Section IV demonstrates the method by applying it to the simulated and experimental data and analyzes the results. Finally, Section V discusses the proposed method and final results and presents a conclusion for this work.

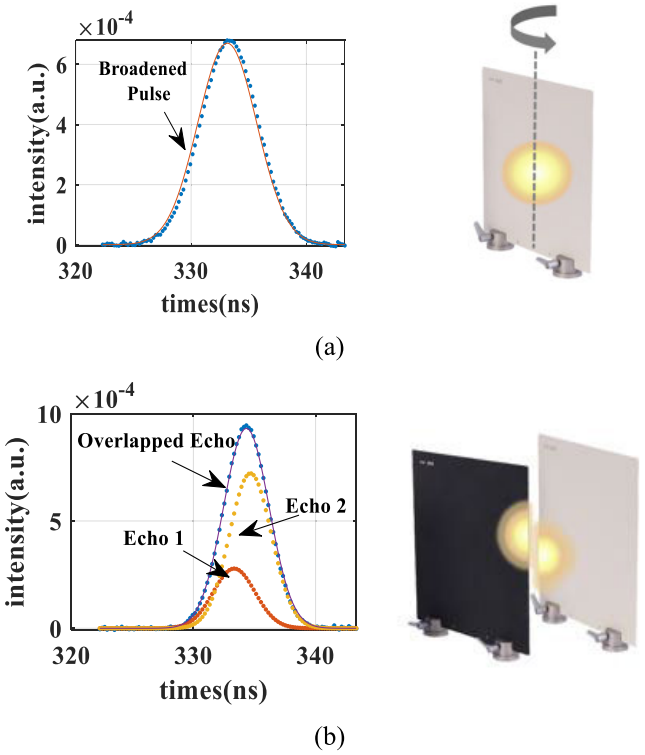


Fig. 1. Diagrams of the two different detecting scenarios obtaining similar echoes that cannot be distinguished intuitively. (a) Broadened echo from a single target. (b) Overlapped echo from two adjacent targets.

II. Methodology

A. Theoretical Analysis

When laser footprint covers neighboring targets at which are close enough, more specifically, the distance between them is less than the range that a single echo counts, for example, 60 cm for 4-ns pulsewidth. The neighboring targets will generate the "pseudo-monopulse." The geometric center position of the "pseudo-monopulse" shifts in different wavelengths due to the targets' different responses in the spectral domain. For example, in a particular wavelength, if the front target has high reflectance, while the latter has lower reflectance, the geometric center position will shift to the front target and vice versa. Therefore, the distribution of geometric center position of the "pseudo-monopulse" shows distinct statistically from that of a single target. The physical nature of this funding lays on that the waveform of HSL can be treated as the accumulation of photons reflected by targets. Thus, the higher amplitude of waveform indicates that more photons are received at that epoch. Also, the change of spatial distribution along the ranging path will introduce the change in the amount of reflected photons, which causes very minor waveform variation in the time domain. However, such variation can be easily detected statistically since its aggregation and asymmetry coincide with the spectral difference of the targets.

To visually show the statistical physical phenomenon that the geometric center position shifts in wavelength caused by spectral factors, we testify our theoretical analysis with both simulation and field tests.

In simulation, we choose three materials with known spectra as the targets on purpose [31], to generate the simulated FWHSL data. redIn Koirala's research, the three materials are goethite, iron, and lime, respectively, which are defined as Material-1, Material-2, and Material-3 in this study. They are set as the targets for general purposes, to show how the spectral difference disturbs the distribution of the center positions of the echo waveforms. The selected criterion is that the spectra of Material-1 and Material-2 are close, while Material-3's spectrum is totally different. Their spectra are shown in Fig. 2(a). For general purposes, we divide the three targets into two groups: Material-1&2 and Material-1&3 to examine the robustness of the proposed method. Material-1 is also chosen as the single target for getting the broadened waveforms by different rotations. We simulate the echo waveforms according to Hu et al.'s research [32], that is, the following conditions hold.

First, we consider that these materials are covered by a small laser footprint and they are extended targets. Actually, they can also be nonextended targets, as long as multiple targets with different spectral reflectance covered by a laser footprint simultaneously.

We mainly focus on the modulation of the incident pulsed laser by the target; we have not considered parameters and coefficients that are not relevant to this modulation, such as the diameter of the receiver optical system.

The impulse response $h_\lambda(t)$ of the laser echo depends on the surface characteristics and is expressed as

$$h_\lambda(t) = \rho_\lambda(\beta) g_\lambda(x, y) \delta\left(t - \frac{2z(x, y)}{c}\right) \tag{1}$$

where λ is the wavelength of the laser, ρ_λ is the bidirectional reflectance distribution function (BRDF) at λ, β_λ is the incident angle, $g_\lambda(x, y)$ is the spatial distribution of the incident laser intensity, δ is the Dirac function, $z(x, y)$ is the distance between the laser transmitter (also laser receiver for the coaxial-line structure of remote sensing equipment) and target, and c is the speed of light.

We consider the modulation of the target on the incident laser waveform as a linear system; the laser echo waveform, $l_\lambda(t)$, is thus the convolution (denoted as the symbol $\bigotimes$ in the following equation) of the impulse response $h_\lambda(t)$ and incident pulse waveform $q_\lambda(t)$:

$$l_\lambda(t) = h_\lambda(t) \bigotimes q_\lambda(t). \tag{2}$$

In this study, both the spatial and time distributions of the incident pulsed laser fit a Gaussian distribution [32], which are expressed as

$$g_\lambda(x, y) = \frac{2P_t}{\pi \omega_\lambda^2} \exp\left(-2\frac{x^2 + y^2}{\omega_\lambda^2}\right) \tag{3}$$

and

$$q(t) = \exp\left(-\frac{t^2}{\tau^2}\right) \tag{4}$$

where τ represents the LiDAR signal width and P_t represents the total signal power. ω_λ is the laser beam radius with different wavelengths λ at point $z(x, y)$ and is expressed as

$$\omega_\lambda = \omega_0 \sqrt{1 + \left(\frac{\lambda z}{\pi \omega_0^2}\right)^2} \tag{5}$$

where ω_0 is the waist radius, $\omega_0 = 2(\lambda/\pi\varphi)$, and φ is the beam divergence angle. Hence, the modulation of the laser and target for a single microfacet is expressed as

$$dl_\lambda(t) = \frac{2P_t \rho_\lambda \cos\beta}{\pi \omega_\lambda^2} \exp\left[-2\frac{x^2 + y^2}{\omega_\lambda^2}\right] \cdot \exp\left\{-\frac{\left[t - \frac{2z(x,y)}{c}\right]^2}{\tau^2}\right\} dx dy. \tag{6}$$

Then, we integrate [32] all the single microfacet $dl_\lambda(t)$ for the extended targets covered be the laser footprint and acquire the modeled echo waveforms.

The following parameters are set for the simulation: 1) the distance between the front target and the HSL is 5 m; 2) the NDs between the two close targets is 20 cm; and 3) the pulsewidth is 4 ns and other, while, for the case of a single-target broadened waveform, the rotation angle is set 60° for a bigger broadened width, to make it more like a "pseudo-monopulse." The schematics of two cases of the broadened echo are shown in Fig. 2(b) and (c). We generate the echoes with 60 wavelengths from 1100 to 1700 nm under these conditions and drew the figures of time wavelength intensity, as shown in Fig. 2(d)–(f). In Fig. 2(d) and (e), the geometric center position of "pseudo-monopulse" has changed dramatically in different wavelengths, while that of the single target's broadened echo seldom changes in Fig. 2(f).

To emphasize the physical difference between the two waveforms, we set thresholds to find outliers based on one standard deviation rule for geometric center position in the spectral domain. We display the figures about the statistics of wavelength-geometric center position in Fig. 2(g)–(i). The geometric center position of the "pseudo-monopulse" shows a distinct distribution, and the outliers of the "pseudo-monopulse" gather in specific wavelength in Fig. 2(g) and (h). In contrast, the geometric center position distribution of a single target is random, and the number of outliers is equally distributed beyond both thresholds, as shown in Fig. 2(i). The significant statistical difference may provide an effective method to detect the "pseudo-monopulse" among others. Also, such method is described in Sections II-B and II-C, while the corresponding experimental verification is demonstrated in Section IV-A.

B. Decision

The proposed method is divided into decision and decomposition, and the details are listed in Fig. 3. In the part of decision, we propose the detailed decision rules of the waveform types. In the part of decomposition, we design a characteristic parameter estimation (CPE) method for more reliable initial waveform parameters and a new waveform decomposition method for HSL.

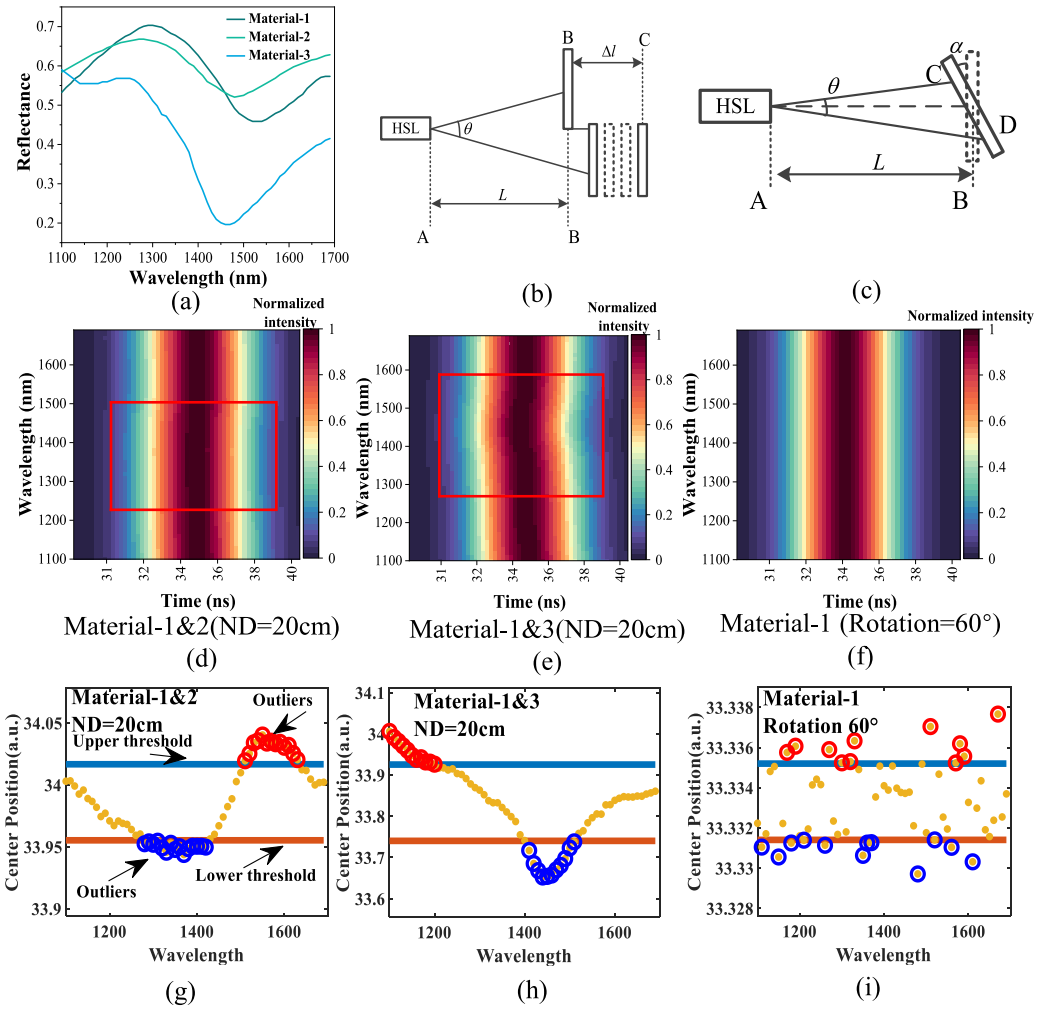


Fig. 2. Theoretical modeling and derivation in determining the waveform types via multiwavelength. (a) Three targets' spectra in waveband from 1100 to 1700 nm. (b), (c), (e), and (f) Center position distributions of the "pseudo-monopulse." (d) and (g) Distribution of center positions of the single target's waveform. (e)–(i) Change of the center position when the SNR is set to 40 dB. The apparent difference in the distribution with wavelength. The red boxes indicate the significant changes in center positions. The red and blue circles are the outlier center positions under different wavelengths.

1) HSL Data Preprocessing: In the HSL's data preprocessing, we perform time-domain synchronization correction in all channels, by calculating the mean value of the transmitted pulses' geometric center position in each channel as the unified starting point [13], [28], as shown in Fig. 4(a) and (b).

Many types of noise contaminate the laser signal during generating, transmitting, interacting, and receiving, including ambient light, photodiode noise, and noise in the amplifier circuit [24], [33]. We adopt a Savitzky–Golay (S–G) filter [24], [33] to smooth the waveform for it can excel in removing noise from the signal while retaining its shape. The window size of the S–G filter is selected as the transmitted pulses' full-width at half-maximum (FWHM) and the polynomial degree is set as three.

2) Waveform Fitting: The two waveform types discussed in this article are all considered as only one single waveform component due to the limited ability of the inflection points in searching for the hidden components [24]. Therefore, one single SND model (7) is adopted to fit the waveform and retrieve the parameters first

$$y_{\lambda_j} = f_{\lambda_j}\left(t|A_{\lambda_j}, s_{\lambda_j}, F_{\lambda_j}, \alpha_{\lambda_j}\right) = A_{\lambda_j}\exp\left[\frac{-(t-s_{\lambda_j})^2}{F_{\lambda_j}^2/(4\ln 2)}\right]\left[1+\mathrm{erf}\left(\alpha_{\lambda_j}\frac{t-s_{\lambda_j}}{F_{\lambda_j}/2\sqrt{\ln 2}}\right)\right] \tag{7}$$

where λ_j represents the wavelength at the jth channel; y_{λ_j} denotes the actual measured waveform data; $f_{\lambda_j}(t)$ denotes the modeled waveform as a function of time t; A_{λ_j}, s_{λ_j}, and F_{λ_j} represent the waveform parameters of intensity, center location, and FWHM, respectively; α_{λ_j} represents the skew coefficient of the SND model; and erf denotes the error function.

We perform the initial parameter estimation on a layer-stripping strategy [20], [28]. Seek the data of the maximum current waveform and its position as the initial intensity and center parameters, and then, determine the FWHM by looking for the half-peak position

$$A_{\lambda_j} = y_{\lambda_j,\max} \tag{8}$$

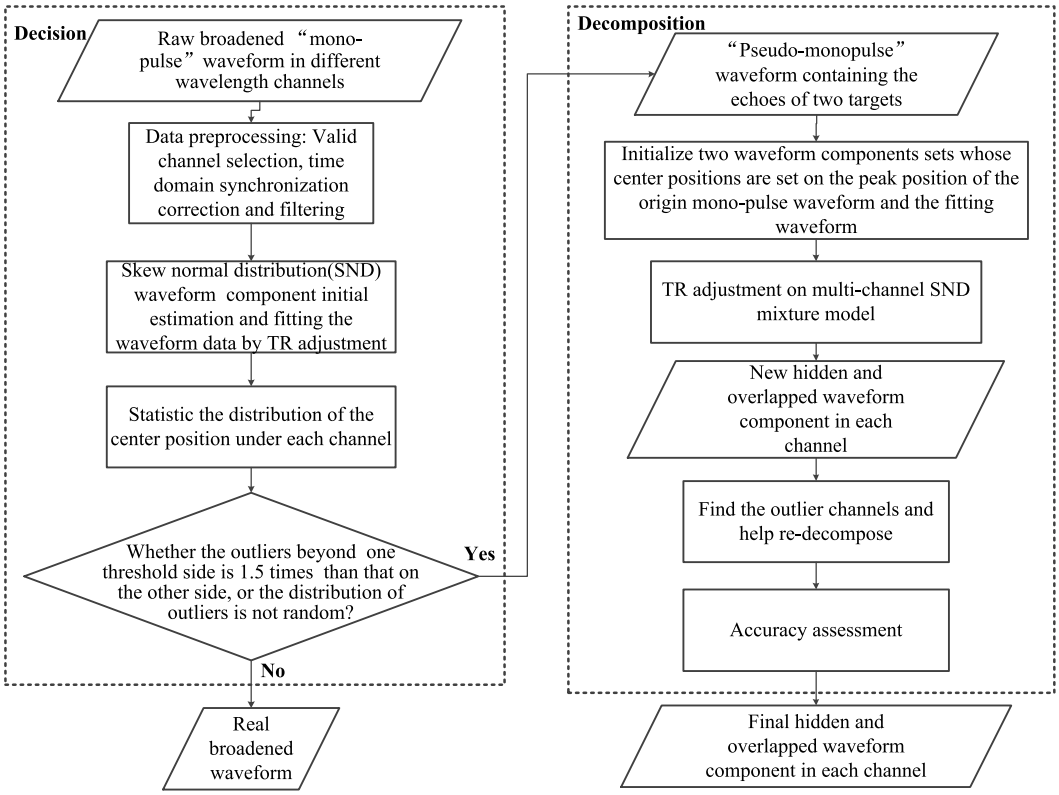


Fig. 3. Flowchart of the range resolution enhancement method, including decision and decomposition.

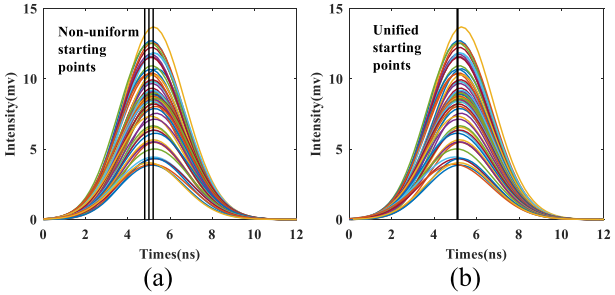


Fig. 4. Time-domain synchronization correction is conducted in order not to affect the TOF measurements. (a) Before time-domain synchronization correction. (b) After time-domain synchronization correction.

$$s_{\lambda_j} = t_{\lambda_j,\max} \tag{9}$$

$$F_{\lambda_j} = t_{\lambda_j,\text{half_max_right}} - t_{\lambda_j,\text{half_max_left}} \tag{10}$$

where $y_{\lambda_j,\max}$ and $t_{\lambda_j,\max}$ denote the maximum value and the corresponding time of the origin waveform, $t_{\lambda_j,\text{half_max_right}}$ and $t_{\lambda_j,\text{half_max_left}}$ represent the left and right half peak position, respectively, and α_{λ_j} is set to 0 at first.

Trust-region (TR) optimization [17], [20], [28] is adopted in the study due to its ability in restraining parameter ranges in advance with prior knowledge. Since the information about the transmitted pulse is known at first, we perform the TR optimization on the SND model using the estimated parameters at each channel, limiting the FWHM range to between 0.8 and 2 times the transmitted pulsewidth and setting the amplitude greater than the noise threshold [17].

3) Statistics of the Geometric Center Position Distribution and Decision: Select the geometric center position parameters in the results of all channels after fitting, calculate the mean value and standard deviation, and set the thresholds for decision, as shown in (8)–(10)

$$\bar{s} = \frac{\sum_{j=1}^{n} s_j}{n_{\text{channels}}} \tag{11}$$

$$\sigma_{\bar{s}_{\text{std}}} = \sqrt{\frac{\sum_{i=1}^{n} (s_i - \bar{s})^2}{n_{\text{channels}}}} \tag{12}$$

$$\text{thr}_{\text{upper}} = \bar{s} + \sigma_{\bar{s}_{\text{std}}}, \quad \text{thr}_{\text{lower}} = \bar{s} - \sigma_{\bar{s}_{\text{std}}}. \tag{13}$$

We observe the geometric center position distribution rules and count the channels whose geometric center position exceeds the thresholds. For the "pseudo-monopulse," its distribution should meet the requirements of "aggregation" or "asymmetry," and the detailed rules are listed as follows.

1) In a certain wavelength, if most channels' geometric center positions exceed the upper or lower threshold and show "aggregation," the echo should be considered as a "pseudo-monopulse" waveform, and it contains two targets' echoes at least.
2) We set an empirical ratio of 1.5, if the number of geometric center positions exceeds the one-side threshold significantly more than the other side (1.5 times); the distribution presents "asymmetry," and the echo is considered as a "pseudo-monopulse" waveform.
3) If the number exceeding the thresholds on both sides does not show obvious "aggregation" or "asymmetry"

(the ratio is less than 1.5), the echo is considered to contain only one target.

Algorithm 1 MCMO

Input:
do
The number of total channels $\leftarrow n$
The number of the outlier channels $\leftarrow a$
Set Serial Number of the channel i $\leftarrow 1$
Set Serial Number of the outlier channel j $\leftarrow 0$
while $i <= n$ **do**
if jth channel is the outlier
$a+=1$
end
end while
for the waveform components of a outlier channel
Set a new initial parameter set
$s_{new,\lambda_j} = \frac{\sum_{i=1}^{n-a} s_{\lambda_{n-a}}}{a}$
$F_{new,\lambda_j} = \frac{\sum_{i=1}^{n-a} F_{\lambda_{n-a}}}{a}$
the parameter A keeps unchanged.
do TR adjustment on the new component sets.
end for
until
$\frac{a}{n} < 10\% \Leftrightarrow \frac{\text{the number(outlier channels)}}{\text{the number(total channels)}} < 10\%$
Output:
re-decomposed subwaveforms and their parameter sets

C. Decomposition

We think that the "pseudo-monopulse" contains two subwaveforms first. As for three targets or more, we will continue to search for relevant characteristics and rules in future studies. We initialize a new waveform component for the "pseudo-monopulse" and reestimate the optimized parameters of the two components. In general, the decomposition is performed in two steps: CPE and multichannel waveform decomposition (MCWD), as shown in the part of decomposition of Fig. 3.

1) Characteristic Parameter Estimation: CPE is specially designed for the "pseudo-monopulse" waveforms utilizing the SND mixture model and is based on the distinction of the three characteristics (geometric center position, FWHM, and skew coefficient) between the echo and fit SND waveform.

When the ND between two targets is relatively close under the "pseudo-monopulse," the peak value increases rapidly, while the pulsewidth ΔF hardly changes, resulting in a sharper waveform shape. Thus, the extended width ΔF is small compared to the transmitted pulse, while the change in skew coefficients $\Delta\alpha$ is large and the ratio $(\Delta F/\Delta\alpha)$ is small. When the ND increases, the waveform's shape will be relatively gentle, resulting in an increased ratio. The ratio was determined by both transmitted pulses and echoes fitting results with the same optimization limitation in TR adjustment.

Therefore, when the ratio $(\Delta F/\Delta\alpha)$ is relatively high, the geometric center position of the first waveform component is considered the midpoint between the left starting point $t_{\lambda_j,\uparrow}$ and the peak position of the "pseudo-monopulse" $t_{\lambda_j,\text{SND_peak}}$. The second component is considered at the peak position of the original waveform data $t_{\text{origin_data_peak}}$. When the ratio is relatively low, two components are located at the peak positions of the original waveform data $t_{\text{origin_data_peak}}$ and the "pseudo-monopulse" $t_{\lambda_j,\text{SND_peak}}$. The initial pulsewidth is set the same as the transmitted pulses. The initial intensity is set to half of the echo's peak value, and the skew coefficients of the two components are set as 0

$$\begin{cases} A_{1,\lambda_j} = \dfrac{y_{\lambda_j,\max}}{2} \\ A_{2,\lambda_j} = \dfrac{y_{\lambda_j,\max}}{2} \end{cases} \tag{14}$$

$$\begin{cases} s_{1,\lambda_j} = \dfrac{t_{\lambda_j,\uparrow} + t_{\lambda_j,\text{SND_half_peak}}}{2} \\ s_{2,\lambda_j} = t_{\lambda_j,\text{origin_data_peak}} \end{cases}$$

$\frac{\Delta F}{\Delta\alpha}$ at a relative high level

$$\begin{cases} s_{1,\lambda_j} = t_{\text{origin_data_peak}} \\ s_{2,\lambda_j} = t_{\text{fitting_data_peak}} \end{cases}$$

$\frac{\Delta F}{\Delta\alpha}$ at a relative low level (15)

$$F_{1,\lambda_j} = F_{2,\lambda_j} = F_{\text{emission}}. \tag{16}$$

The parameter ranges are set to assist the TR optimization and prevent the overfitting from causing negative values during the optimization [17]. When the ratio is relatively high, for the first component, the upper limitation for the geometric center position $s_{1,\lambda_j,\text{upper}}$ is set at the midpoint between the SND model's left half peak $t_{\lambda_j,\text{SND_half_peak}}$ and its geometric center position $t_{\lambda_j,\text{SND_peak}}$, while the lower limit $s_{1,\lambda_j,\text{lower}}$ is set at the signal's rising edge $t_{\lambda_j,\uparrow}$. As for the second component, the upper $s_{2,\lambda_j,\text{upper}}$ and lower limits $s_{2,\lambda_j,\text{lower}}$ are set at the midpoint between the geometric center positions of the SND model and the original waveform data, and the signal's falling edge $t_{\lambda_j,\downarrow}$, respectively. At a low ratio level, both components' upper and lower limits are set as the mean value of the geometric center position counted in all channels plus or minus its three times standard deviation $s_{i,\lambda_j,\text{lower}} \in [s_{i,\lambda_j} - 3\sigma_{\text{SND}}, s_{i,\lambda_j} + 3\sigma_{\text{SND}}]$. The ranges are listed in (17), including the intensity and FWHM.

Parameter ranges on the scope of TR optimization

$\frac{\Delta F}{\Delta\alpha}$ at a relative high level

$$\begin{cases} A_{1,\lambda_j,\text{lower}} = 0, \quad A_{2,\lambda_j,\text{lower}} = 0 \\ s_{1,\lambda_j,\text{lower}} = t_{\lambda_j,\uparrow} \\ s_{1,\lambda_j,\text{upper}} = \dfrac{t_{\lambda_j,\text{SND_half_peak_left}} + t_{\lambda_j,\text{SND_peak}}}{2} \\ s_{2,\lambda_j,\text{lower}} = \dfrac{t_{\lambda_j,\text{SND_peak}} + t_{\lambda_j,\text{origin_data_peak}}}{2} \\ s_{2,\lambda_j,\text{upper}} = t_{\lambda_j,\downarrow} \\ F_{1,\lambda_j,\text{lower}} = F_{2,\lambda_j,\text{lower}} = F_{\text{emission}} \end{cases}$$

$\frac{\Delta F}{\Delta\alpha}$ at a relative low level

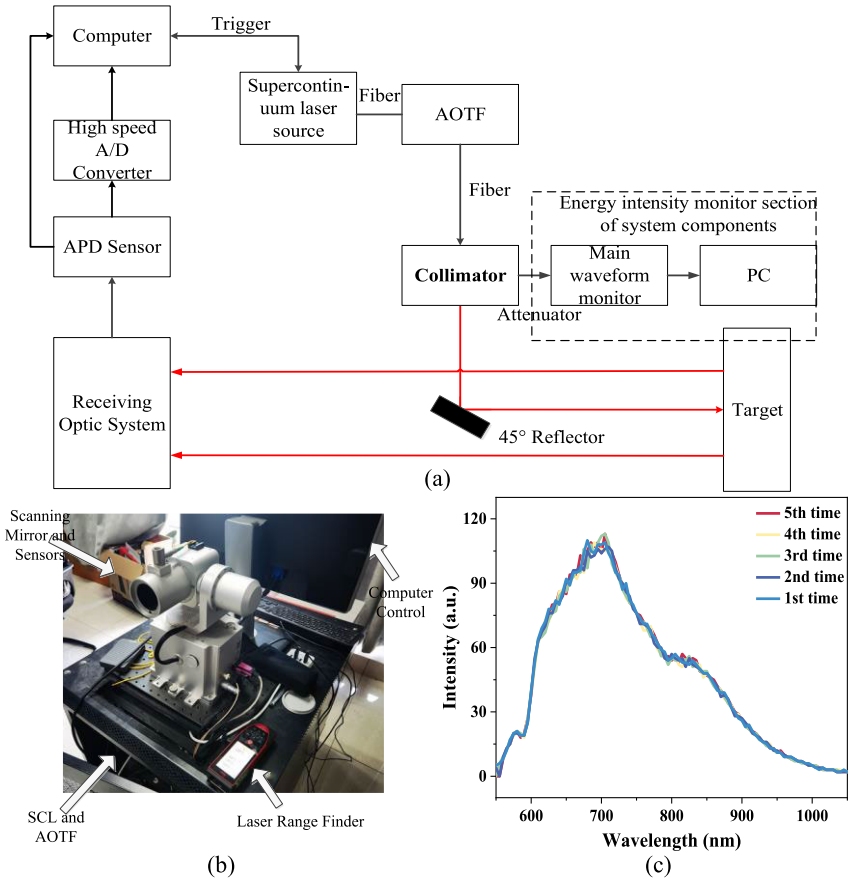


Fig. 5. HSL system used in the experiment. (a) and (b) Structure and the prototype of the HSL system. (c) HSL system energy stability test results.

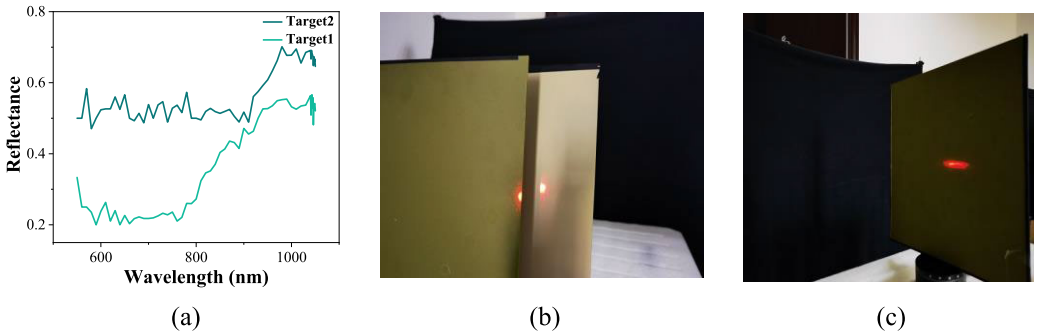


Fig. 6. (a) Two boards' spectra in wavelength from 550 to 1050 nm. Experimental setup was used to obtain the laser echoes scattered from (b) two close targets with varying intervals and (c) single target with varying rotation angles.

$$\begin{cases} A_{1,\lambda_j,\text{lower}} = 0, \quad A_{2,\lambda_j,\text{lower}} = 0 \\ s_{1,\lambda_j,\text{lower}} = s_{1,\lambda_j} - 3\sigma_{\text{SND}}, \ s_{1,\lambda_j,\text{upper}} = s_{1,\lambda_j} + 3\sigma_{\text{SND}} \\ s_{2,\lambda_j,\text{lower}} = s_{2,\lambda_j} - 3\sigma_{\text{SND}}, \ s_{2,\lambda_j,\text{upper}} = s_{2,\lambda_j} + 3\sigma_{\text{SND}} \\ F_{1,\lambda_j,\text{lower}} = F_{2,\lambda_j,\text{lower}} = F_{\text{emission}}. \end{cases} \tag{17}$$

2) Multichannel Waveform Decomposition: MCMD includes single-channel waveform decomposition (SCWD) and multichannel mutual optimization (MCMO). In SCWD, we replace the parameters in the SND mixture models with the previously estimated parameters and perform TR optimization. We evaluate the accuracy of the waveform decomposition results through three widely used statistical estimators [30], coefficient of determination ($R2$), root-mean-square error (RMSE), and relative RMSE (rRMSE). Meanwhile, based on the relative ND error proposed by Wang et al. [13] and Song et al. [30], to take the decomposition results in all channels into account, we propose the relative error of overall average distance (REOA) of multiple channels to evaluate the progress of the ranging error [28]

$$\text{RMSE}_{\lambda_j} = \sqrt{\frac{1}{n}\sum_{i=1}^{n}\left(y_{\lambda_j} - f_{\lambda_j,i}(t)\right)} \tag{18}$$

$$\text{rRMSE}_{\lambda_j} = \frac{\text{RMSE}_{\lambda_j}}{\bar{y}_{\lambda_j}} \tag{19}$$

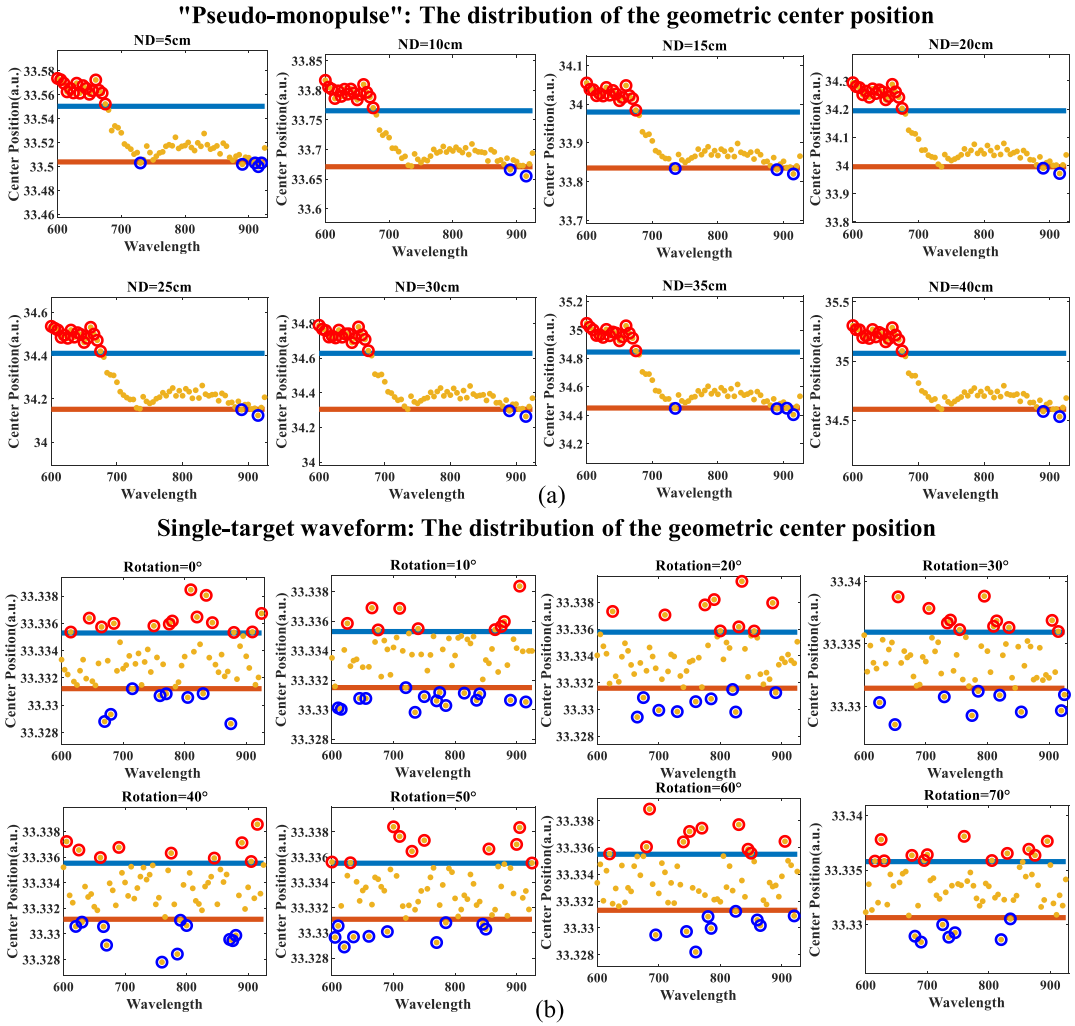


Fig. 7. Statistics of two simulated waveforms' geometric center position distributions in wavelength from 600 to 925 nm at (a) eight kinds of NDs and (b) eight different rotations. ND indicates the neighbor distance of the two targets. Rotation indicates the target's inclination angle.

$$R^2_{\lambda_j} = 1 - \frac{\sum_{i=1}^{n} \left(y_{\lambda_j} - f_{\lambda_j,i}(t)\right)^2}{\sum_{i=1}^{n} \left(y_{\lambda_j} - \bar{y}_{\lambda_j}\right)^2} \tag{20}$$

$$\text{REOA} = \frac{\left|\bar{s} - s_{\text{real}}\right|}{s_{\text{real}}} \tag{21}$$

where $y_{\lambda_j,i}$ and $f_{\lambda_j,i}(t)$ denote the measured value and estimated value at the jth channel, $\bar{y}_{\lambda_j}$ denotes the mean value of the measured echo waveform, and $\bar{s}$ and s_{real} represent the mean value of measured and the actual NDs in all channels, respectively.

Some channels' decomposition results are unreliable due to the rough initial estimation after SCWD. Therefore, we perform MCMO by calculating the mean value of all measured NDs and its standard deviation, and setting the thresholds using (11)–(13) to find the outlier channels exceeding the thresholds. For the channels within the thresholds, the mean value of these geometric center positions and FWHM are considered as the right and new initial parameters. The limitation of the optimization range remains unchanged. Keep looking for outliers and help them redecompose until they make up less than 10% in all channels. The procedure is presented as the following pseudocode.

III. Materials

We design an experiment to verify the theoretical analysis. It is expected to be widely used in multitarget and close-range scenarios, such as hidden target detection and canopy vegetation investigation.

A. HSL System and Experiments

The structural diagram and real prototype of the HSL system are shown in Fig. 5(a) and (b). The HSL system consists of a supercontinuum laser (SCL) source red(SC-OEM-HP, YSL), an acoustic-optical tunable filter (AOTF), and a data-collection computer (including a customized 5-GHz/s high-speed acquisition card) [16], [34]. A commercial laser range finder (DELI Laser Rangefinder DL4171) confirms the distance.

The operational spectrum of the system is from 550 to 1050 nm with a maximum power of more than 8 μJ and 5-nm spectral resolution. The pulse repetition rate is 500 kHz on average, and the average FWHM is 4 ns [16], [34]. The scattered laser pulses are collected by a Cassegrain telescope (with 700-mm focal length and 100-mm aperture diameter).

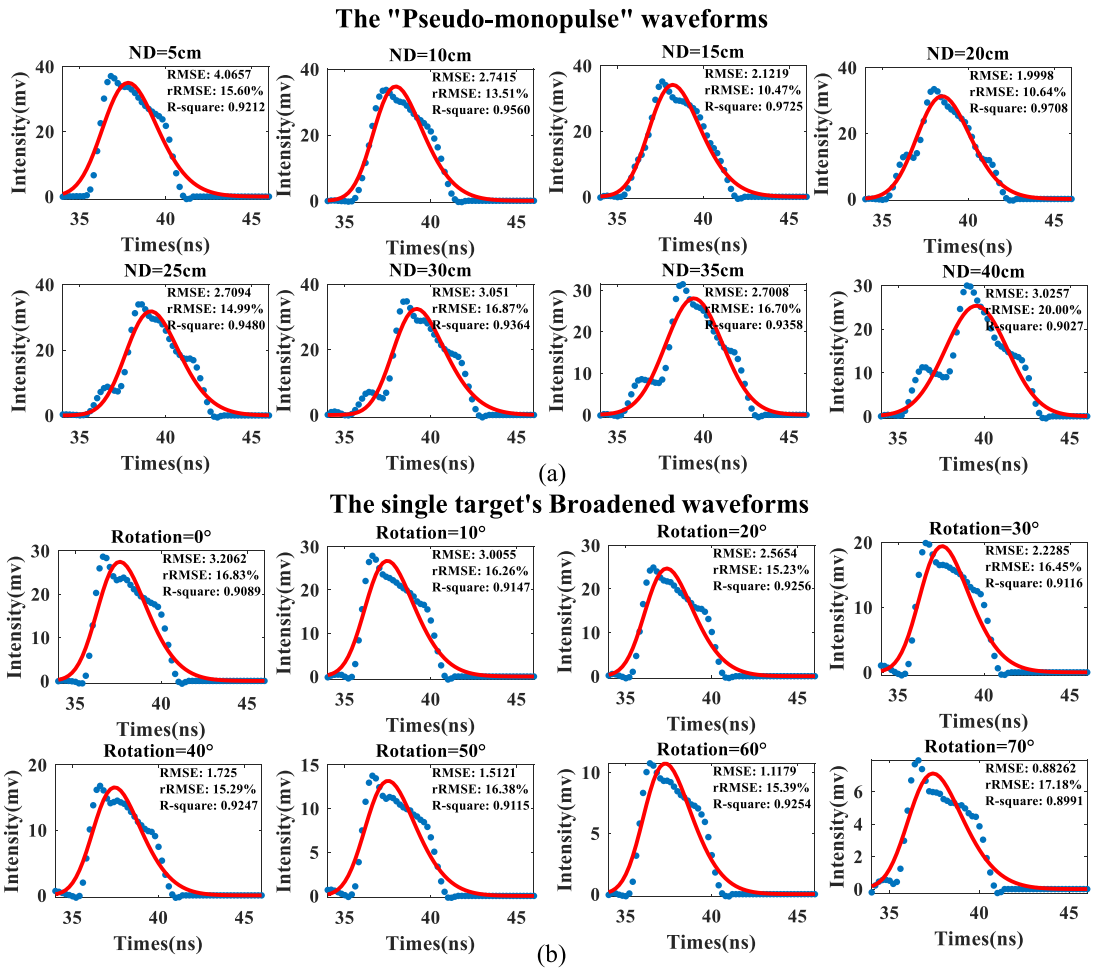


Fig. 8. Comparison of (a) "pseudo-monopulse" waveforms and (b) single target's broadened echoes. The blue dots are the actual echoes, and the red lines are the fitting results.

The focal point of the telescope is imaged onto a high-voltage-biased low-noise-level avalanche photodiode (APD) module [9] consisting of a silicon APD and an integrated amplifier with a bandwidth of 1 GHz, which converts the laser echo into an electronic signal and amplifies it, effectively suppressing the noise. After testing, the system could obtain hyperspectral waveform data at 37.5 m [9].

AOTF [9] is a tunable narrow bandpass filter with bandwidths ranging from a few to tens of nanometers, which exploits acousto-optic effects to diffract and shift the frequency of light. In the proposed system, due to the high tuning speed of the AOTF, it only takes half a millisecond to obtain the whole spectrum in the current full spectrum configuration when the AOTF device is synchronized with the broadband laser source.

To verify the data stability, we made several measurements of a standard reflectance board with 99% reflectivity. It can be observed from Fig. 5(c) that the five measured spectra of the board coincide with each other and are always in a stable range, which shows the stability of the employed laser source. The SCL has the power monitors (using a main waveform monitor and a data-collection PC) as a black dotted box shown in Fig. 5(a). There is a main wave detector in the system by splitting emitted laser beams that can monitor energy stability.

We design two experiments to explore the influence of spectral properties on the geometric center position distribution of the two waveform types. First, HSL measures the two close neighboring targets at a distance of 5 m. The distance between the two targets is set successively from 5 to 40 cm with a step of 5 cm. Based the on available literature [13], [28], the best achievable range resolution is 40 cm with current methods, considering a 4-ns pulsewidth. The laser footprint covers two targets simultaneously, while for the second experiment, we remove the latter target and rotate the front target from 0° to 70° in a step of 10°, keeping other conditions unchanged, to obtain the echo waveform from the single target. The experiments are shown in Fig. 6(b) and (c). One issue that should be specifically addressed is that the two test boards are made of different materials in the experiment, and thus, their reflectance is distinctly different.

B. Datasets

Two datasets are used in our study: simulated FWHSL echoes to evaluate the efficiency, robustness, and feasibility of the proposed method, and a measured dataset from FWHSL to assess the decision rules and decomposition accuracy of the proposed method and explore its ability to identify the adjacent targets.

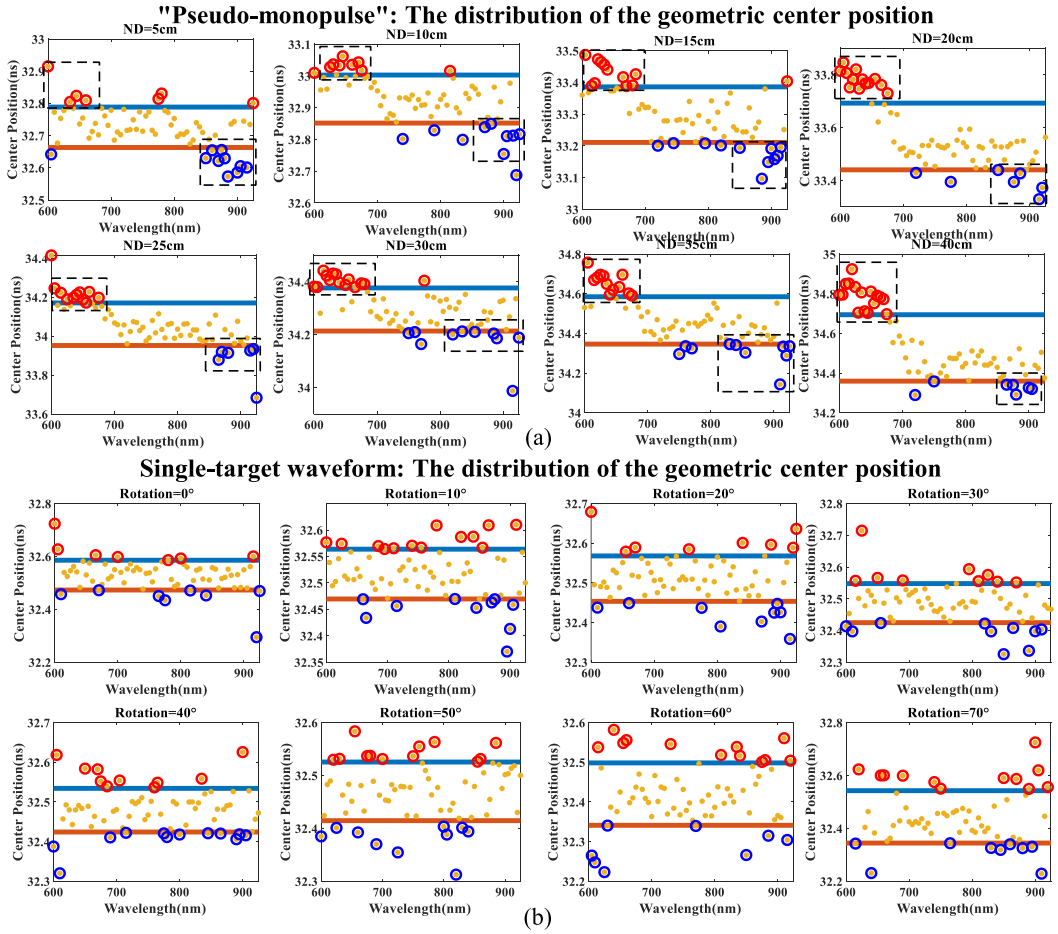


Fig. 9. Statistics of the geometric center position distributions of (a) "pseudo-monopulse" and (b) single target's broadened waveforms in channels from 600 to 925 nm. In each subfigure, the yellow dots represent the geometric center position of the SND model in each channel, the light blue and brown lines represent the calculated thresholds, and the red and blue circles indicate the outliers.

TABLE I

STATISTICS OF THE OUTLIERS AND FIRST MOMENT OF THE GEOMETRIC CENTER POSITION DISTRIBUTION OF THE TWO WAVEFORM TYPES

True neighbor distance(cm)	Mean value of the position(ns)	Standard deviation of the position(ns)	Number above the upper threshold	Number below the lower threshold	Rotation (°)	Mean value of the position(ns)	Standard deviation of the position(ns)	Number above the upper threshold	Number below the lower threshold
5	32.726	0.0628	7	10	0	32.5294	0.0563	7	8
10	32.9275	0.0757	9	10	10	32.5167	0.0471	13	10
15	33.2987	0.0876	12	11	20	32.5108	0.0573	8	9
20	33.5666	0.1261	13	7	30	32.4863	0.0615	9	10
25	34.0613	0.1092	11	6	40	32.4792	0.055	10	12
30	34.2953	0.0815	14	10	50	32.47	0.0554	12	10
35	34.4658	0.1194	12	10	60	32.4198	0.0787	12	8
40	34.5271	0.1674	16	7	70	32.4436	0.099	12	9

1) Simulated Data: We measured the true spectral reflectance of the two target boards from in wavelength (550–1050 nm) as a comparison to experimental data in advance. We generate the simulated FWHSL echoes and count the geometric center position distribution. Two targets' spectra are shown in Fig. 6(a). The digital time sampling rate is set to 5 G/s same as the FWHSL system.

2) Experimental Data: We obtain the two groups of echo waveforms both from dual close targets and single target as described in Section III-B and evaluate the proposed method.

IV. RESULTS

A. Simulation and Experimental Data for Decision of the Waveform Type

1) Simulated Data: We simulate and count the geometric center position distributions of the "pseudo-monopulse" and the single-target broadened waveforms in the wavelength from 600 to 925 nm, as shown in Fig. 7(a) and (b). In Fig. 7(a), the outliers of the "pseudo-monopulse" show aggregation and asymmetry in all NDs and significantly differ from the random distribution from a single target with all rotations, as shown in Fig. 7(b).

TABLE II
STATISTICS OF THE MEAN VALUE OF THE INITIAL AND BROADENED WIDTH, AND THE CHANGES OF THE SKEW COEFFICIENTS

True neighbor distance(cm)	Transmitted pulse Mean value FWHM(ns)	α	Echo Mean value FWHM(ns)	α	ΔF	$\Delta\alpha$	ΔF/$\Delta\alpha$
5	4.2385	0.8077	5.1171	1.6476	0.8786	0.8399	1.0461
10	4.2492	0.8032	5.1506	0.9014	0.9014	0.7707	1.1697
15	4.2465	0.813	5.3693	1.5312	1.1228	0.7182	1.5635
20	4.2339	0.7933	5.5178	1.4942	1.2839	0.7009	1.8319
25	4.2439	0.8013	5.5138	1.4727	1.27	0.6714	1.8914
30	4.2532	0.8078	5.5533	1.4366	1.3001	0.6288	2.0676
35	4.2301	0.784	5.6435	1.4032	1.4134	0.6192	2.2829
40	4.2562	0.8184	5.9095	1.3145	1.6533	0.4961	3.3322

TABLE III
MEAN VALUE OF THE INITIAL PARAMETERS AND RANGES ESTIMATED IN DIFFERENT NEIGHBOR DISTANCES

True neighbor distance(cm)		5		10		15		20	
Component		Estimate	Bound	Estimate	Bound	Estimate	Bound	Estimate	Bound
1	A_1	$A/2$	[0,A]	$A/2$	[0,A]	$A/2$	[0,A]	$A/2$	[0,A]
	F_1	F_0	[4,5]	F_0	[4,5]	F_0	[4,5]	F_0	[4,5]
	s_1	36.65	[36.30,36.99]	36.84	[36.49,37.20]	36.5	[35.8,36.8]	36.4	[35.8,37.3]
	α_1	1	[0,2]	1	[0,2]	1	[0,2]	1	[0,2]
2	A_2	$A/2$	[0,A]	$A/2$	[0,A]	$A/2$	[0,A]	$A/2$	[0,A]
	F_2	F_0	[4,5]	F_0	[4,5]	F_0	[4,5]	F_0	[4,5]
	s_2	36.83	[37.17,36.48]	37.27	[36.91,37.63]	37.6	[37.7,41.6]	38	[37.7,41.8]
	α_2	1	[0,2]	1	[0,2]	1	[0,2]	1	[0,2]

True neighbor distance(cm)		25		30		35		40	
Component		Estimate	Bound	Estimate	Bound	Estimate	Bound	Estimate	Bound
1	A_1	$A/2$	[0,A]	$A/2$	[0,A]	$A/2$	[0,A]	$A/2$	[0,A]
	F_1	F_0	[4,5]	F_0	[4,5]	F_0	[4,5]	F_0	[4,5]
	s_1	36.4	[35.8,37.3]	36.2	[35.2,37.7]	36.1	[35.2,37.4]	36	[35.2,37.5]
	α_1	1	[0,2]	1	[0,2]	1	[0,2]	1	[0,2]
2	A_2	$A/2$	[0,A]	$A/2$	[0,A]	$A/2$	[0,A]	$A/2$	[0,A]
	F_2	F_0	[4,5]	F_0	[4,5]	F_0	[4,5]	F_0	[4,5]
	s_2	38.4	[37.8,42.2]	38.6	[38.0,42.6]	38.8	[38.1,42.8]	39.2	[38.2,42.8]
	α_2	1	[0,2]	1	[0,2]	1	[0,2]	1	[0,2]

2) Experimental Data: We perform the experiment and obtain the waveforms by the HSL within 66 channels from 600 to 925 nm. We display the echo waveform data in 700 nm and the fitting results by a single SND model and using (8)–(10) in two cases, as shown in Fig. 8(a) and (b). One single SND model can intuitively fit these two types of waveforms well. From the accuracy assessment, RMSE and rRMSE are all at a low level. $R2$ exceeds 0.9 in most cases except that the rotation angle is more than or equal to 70°. Thus, the fitting results by a single SND model are reliable in the two cases.

When the NDs are 5, 10, and 15 cm, the waveforms are quite similar to the broadened waveform when the rotation angle is less than 70°. It is difficult to distinguish these two waveform types without prior knowledge. Since the laser signal generated by the HSL system is not the standard Gaussian or SND distribution, the two spikes in Fig. 8(a) roughly reveal the corresponding two targets' waveform components when the ND is more than 20 cm. Although inflection points can detect these components, they may be abandoned later for not meeting the pulsewidth condition, which should be no less than the transmitted pulse [30].

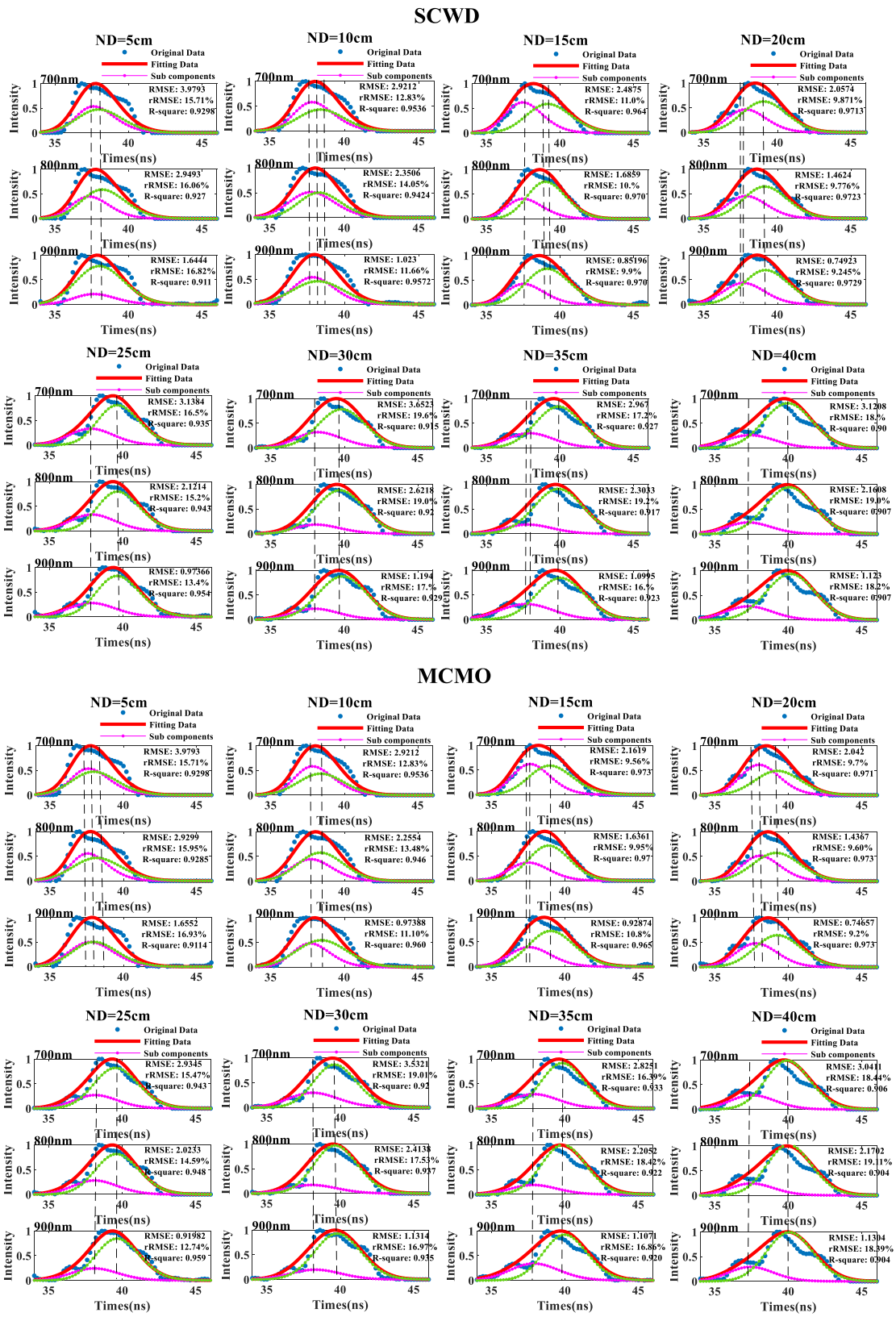


Fig. 10. "Pseudo-monopulse" echoes at 700, 800, and 900 nm were decomposed by SCWD and MCMO. The blue dots represent the measured echo waveform, the red lines represent the fit waveform, and the pink and green dotted lines represent the decomposed subwaveform components.

The geometric center position distributions of these two types are shown in Fig. 9. In Fig. 9(a), when the ND changes from 5 to 40 cm in a step of 5 cm, the outliers exceeding the thresholds exist in the wavelength from 600 to 680 nm and from 850 to 925 nm. The geometric center position distributions of the "pseudo-monopulse" always present the

TABLE IV
STATISTIC OF ALL DECOMPOSITION RESULTS AND ACCURACY ASSESSMENT IN AVERAGE FROM ALL CHANNELS AT EIGHT DIFFERENT NEIGHBOR DISTANCES, FROM 5 TO 40 cm

Neighbor distance(cm)	Method	Target 1		Target 2		Neighbor distance(cm)	REOA	RMSE	rRMSE	R^2
		Mean value of position (cm)	Standard deviation of position (cm)	Mean value of position (cm)	Standard deviation of position (cm)					
5	SCWD	484.0135	1.8519	488.9696	2.2218	5.13±2.53	0.026	2.8593	16.30%	0.924
	MCMO	484.478	1.5262	489.6865	2.0167	5.32±1.31	0.064	2.9083	16.50%	0.921
10	SCWD	485.8655	2.0385	495.6927	2.2584	9.83±2.98	0.017	2.049	13.10%	0.95
	MCMO	486.1597	1.6514	496.3295	2.0205	10.17±1.66	0.017	2.0505	13.10%	0.949
15	SCWD	482.692	1.4467	505.5937	2.2774	22.9±1.56	0.527	1.6433	10.80%	0.965
	MCMO	483.3322	1.8556	505.4509	1.9435	22.12±1.42	0.475	1.569	10.40%	0.968
20	SCWD	485.4107	2.2383	509.3704	1.3749	23.96±2.27	0.198	1.4811	10.60%	0.968
	MCMO	485.7483	2.2063	509.5846	1.4011	23.84±1.89	0.192	1.4705	10.50%	0.968
25	SCWD	487.2659	2.1969	515.9249	3.2302	28.67±3.14	0.147	2.3966	18.40%	0.919
	MCMO	487.9264	2.4073	515.5744	2.6289	27.65±2.41	0.106	2.3074	17.60%	0.926
30	SCWD	490.9376	3.6091	522.1086	2.8568	31.17±3.01	0.039	2.9355	23.30%	0.888
	MCMO	492.2017	3.4861	522.2897	2.659	30.09±2.57	0.003	2.8392	22.30%	0.898
35	SCWD	486.6945	3.8514	524.9819	2.884	38.29±3.25	0.094	2.6811	23.30%	0.878
	MCMO	489.3871	4.8931	525.1738	2.6072	35.79±3.58	0.023	2.4568	21.60%	0.895
40	SCWD	484.5227	3.5091	526.2514	3.0698	41.73±2.63	0.043	2.3859	22.50%	0.878
	MCMO	484.9813	3.1152	525.9131	2.657	40.93±1.95	0.023	2.2839	21.60%	0.887

"aggregation" and "asymmetry." Compared to the simulated statistical distribution in Fig. 7(a), they are similar and both different from that of a single-target broadened waveforms. According to the spectra of the two targets in Fig. 6(a), the front target responds weaker than the latter in the wavelengths from 600 to 680 nm, so the geometric center position shifts to the latter at first. In the wavelength from 850 to 925 nm, two targets respond similarly, so the geometric center position shifts to the former.

As shown in Fig. 9(b), the single target's geometric center position distributions are random and symmetrical, regardless of how the rotation angle changes. We list the statistics of the outlier counts and the first moment about these two types of waveforms in Table I. For the single target's broadened waveforms, the outliers exceeding one side threshold are roughly equal to the other side, which is obviously different from the shifted geometric center positions of the "pseudo-monopulse" in specific wavelengths.

Both simulated and experimental data verify the previous theoretical analysis. Spectral factor seldom affects the geometric center position distributions of a single target's echoes. However, for the "pseudo-monopulse," due to the different spectral responses of two targets to the same wavelength, the geometric center position will shift to one target and its distribution will show aggregation and asymmetry. The hyperspectral properties can help distinguish two waveform types and the method described in Section II-B is effective for all NDs in the experiment. Table I The statistics of the outliers and first moment of the geometric center position distribution of the two waveform types.

B. Waveform Decomposition Results

Table II shows the statistics of the mean values of the fitting results by a single SND model in wavelengths (600–925 nm) about the transmitted pulses and the echoes at different NDs, and the variation of the pulsewidth and the skew coefficients. When the ND is 5 and 10 cm, the ratio $(\Delta F/\Delta\alpha)$ is low and close to 1. When the ND varies between 15 and 30 cm, the ratio quickly increases to the interval between 1.5 and 1.9. When the ND is more than 30 cm, the ratio grows to more than 2.

We think that the ratio below 1.5 belongs to a relatively low level and the ratio above 1.5 belongs to a relatively high, treating the different waveforms with the corresponding initial parameter estimation methods.

The estimation for parameters and their ranges at the different NDs is listed in Table III. A represents the peak value

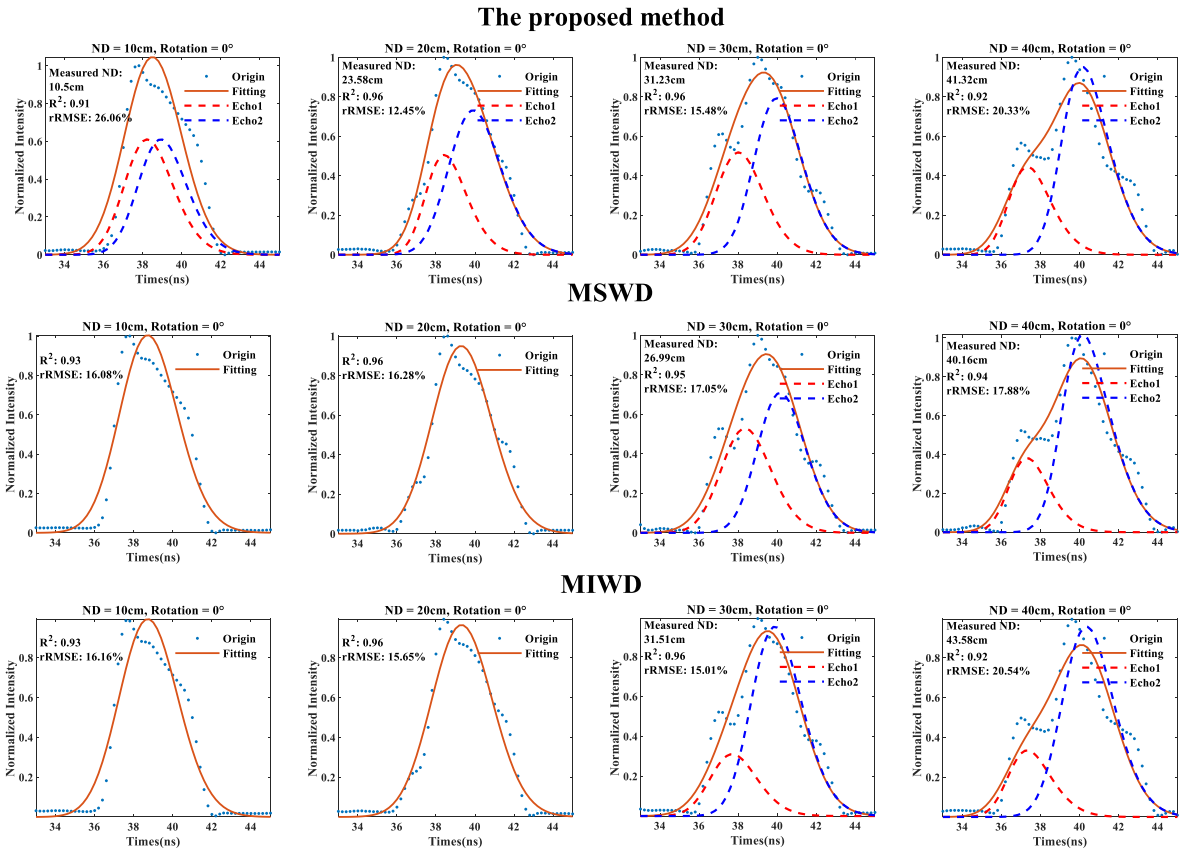


Fig. 11. Comparison of the three waveform decomposition methods. (Top) Proposed method. (Middle) MSWD method. (Bottom) MIWD method.

of the measured waveform, F_0 represents the pulsewidth of the transmitted pulse, and "[,]" indicates the ranges for a certain parameter. When the ND is 5, 10, and 15 cm, the estimation of the position parameters is close to the actual ND. When the ND is between 20 and 30 cm, an error exists less than 5 cm. When it is greater than or equal to 30 cm, an error exists within 10 cm.

The SCWD and MCMO are performed in order to solve the problem that the same component's position varies in different channels. We display the waveform decomposition results with normalized intensity in three channels (700, 800, and 900 nm) in Fig. 10. After MCMO, the variation of the geometric center positions in all channels is corrected except that ND is 15 and 20 cm.

We display the statistic of all decomposition results and accuracy assessment in average from all channels at eight different NDs in Table IV. The symbol " " in Table IV means the maximum range of the measured ND. The number behind " " is the standard deviation of the measured NDs in all channels.

In Table IV, we display the decomposition results by the proposed method and their accuracy assessment results using (18)–(21). When the ND is 5 and 10 cm and from 25 to 40 cm, the overlapped waveforms are well decomposed by the proposed method (REOA and SCWD: 0.043–0.147; MCMO: 0.023–0.106). The minimum of the REOA corresponding to 5 and 10 cm is only 0.026 and 0.017, indicating that the maximum error is not more than 0.13 or 0.17 cm, respectively. In MCMO, the overall decomposition accuracy is improved by providing more reliable parameters based on the decomposition results within the thresholds. However, when the ND is 15 and 20 cm, the results are not accurate and the error is relatively large (REOA and SCWD: 0.198–0.527; MCMO: 0.192–0.475); the final decomposition results are not satisfactory enough. There remain better parameter estimation and optimization methods to be explored.

In addition, SCWD and MCMO are extremely dependent on the accuracy of initial estimation of parameters and their ranges. Without the accurate initial parameter, the decomposition results in different wavelengths will vary significantly, and the standard deviation of the NDs will be large. In that case, the decomposition results are not reliable.

C. Waveform Decomposition Method Comparison

We compare the proposed method with the multispectral waveform decomposition (MSWD) method (proposed by Song et al. [30]), multichannel interconnection waveform decomposition (MIWD) method (proposed by Wang et al. [13]). The waveform decomposition results conducted by the three methods are visually shown in Fig. 11, under the conditions with the interval of 10, 20, 30, and 40 cm.

For MSWD [30], we present the decomposition results of the channel at 700 nm, which recorded the best data in the HSL system. It detects hidden components by comparing whether the center position of the same waveform component between different channels is greater than half of the FWHM. When

it faces such highly overlapped echo waveforms, it performs unsatisfactorily when the interval decreases to 10 and 20 cm and has some error in ranging and ranging resolution compared to the proposed method when the interval is 30 and 40 cm. It is hard for MSWD to reveal hidden components when the interval is below or equal to 40 cm, for a 4-ns pulse (60 cm), due to the low intensity and little center position shifts (less than 2 ns, 30 cm) in different channels of the components. Therefore, the range resolution improvement for MSWD is limited by its working principles.

For MIWD [13], it has utilized the waveforms in multichannels with multiecho quality (MEQ) [13] to obtain an accumulated echo with higher SNR and deformation, which is helpful for the estimation. However, MIWD cannot know whether the waveform contains multiple targets compared to the proposed method. MIWD performs unsatisfactorily if there is no deformation after accumulation. In Fig. 11, MIWD can perform well with the accumulated waveforms whose intervals are 30 and 40 cm and whose rotation angles are low, but there are still some errors compared to the proposed method. When the interval decreases to 10 and 20 cm, leading to litter deformation of the accumulated waveform, MIWD cannot reveal two components accurately.

Comparing the three methods, the proposed method has the smallest error in obtaining the spatial information from the footprint and the highest range resolution. This indicates that the proposed method performs better in revealing hidden components and the improvement for the waveform decomposition accuracy than the current prevailing methods.

V. Conclusion and Discussion

In this study, the echo data of FWHSL are used statistically for the first time to decide the type of echoes and to judge whether it contains multitarget. Then, we design a hyperspectral waveform decomposition method for the "pseudo-monopulse," with which the current methods cannot well deal. The previous studies [35], [36] about the HSL have focused on the spectral detection capability by increasing receiving channels. Unlike in the past, this article proves that the hyperspectral properties extracted from FWHSL measurements can increase the range resolution. For a 4-ns width laser pulse, the range resolution can be boosted to 40 cm by the current most advanced single-wavelength [24], multispectral [30], and hyperspectral [13] waveform decomposition methods. However, none of them are able to well decompose the echo waveform when the neighbor target overlapping with more than 90%. Through the proposed decision method, we observe the apparent difference of statistical geometric center point between "pseudo-monopulse" and the single-target echo waveform. Also, we can tell whether the echo waveform contains multiple targets.

After that, to decompose the "pseudo-monopulse," we give two different initial parameters and their ranges estimation methods according to the ratio of the broadened pulsewidth and the change of the skew coefficient. Then, we use the TR for the optimization. The decomposition results are satisfactory in general after performing MCMO. Also, we find that the initial estimation for parameters and their ranges should be extremely accurate. Through the decomposition method, the "pseudo-monopulse" can be effectively decomposed even when the distance between the two targets reduces to 5 cm at most. However, the proposed method is not ideal for the echo waveform when the ND is 15 and 20 cm, which needs further improvement. This is because of the nonideal Gaussian waveform emitted by the FWHSL and the limitations of the principles of the TR algorithm. It would be better to improve the optimization algorithm or update the FWHSL system in the future.

Finally, for the FWHSL waveform data, there are still more features to be studied and discovered to help develop more accurate waveform processing methods. In addition, the range resolution is still limited by the digital sampling rate of the digitizer system, which is 5 G/s in this test. The difference will not be discovered if the ND is below 3 cm. The proposed method can also be integrated into the FWHSL system to provide more accurate knowledge to facilitate the point cloud's expansion, marking, and other detecting transactions.

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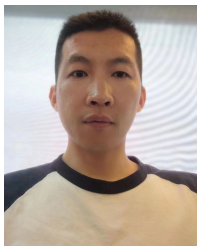
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